

KINETIC THEORY OF GLASSES

Molecular Nature of Matter

- many scientists have proposed the atomic hypothesis. According to this theory, atoms make up everything in the universe.
- The atomic theory of Dalton is also known as the molecular theory of matter. This theory establishes that matter is composed of molecules, which are composed of atoms.

Solids, liquids and gases in terms of molecular Structure

Basis of differences	Solids	Liquids	Gases
The Inter Atomic Distance or the Distance between the molecules	The molecules in solids are very tightly packed and the inter-atomic distance is minimum	The liquid molecules are less tightly packed and the inter-atomic distance between the liquid molecules is more than the Solids	The gas molecules are loosely packed. The gas molecules move freely in the space and the inter-atomic distance is minimum in gas particle.
The average distance a molecule can travel without colliding is known as the mean free path	Solids have no mean free path	Less mean free path than gases	A mean free path is present in gas molecules

Behaviour of Gases

- Gases at low pressure and temperatures much higher than those at which they liquify (or solidify) approximate a relationship between pressure, temperature and volume.

$$PV = KT$$

This is the universal relationship that all gases satisfy.

➤ Avagadro's Hypothesis

- At constant temperature and pressure, equal volume of all gases contain equal number of molecules.
($V \propto \text{no. of molecules}$)

Where P , V and T denote pressure, volume and temperature, and k denotes the constant for a given volume of gas.

$$k = NK_B$$

where N is the number of molecules, k_B is the Boltzmann constant whose value never changes

$$PV = Nk_B T$$

- Therefore $\frac{PV}{NT} = k_B$, this will be same for all gases
- Two gases having pressure, temperature and volume as (P_1, T_1, V_1) and (P_2, T_2, V_2) are considered.
- Therefore, $\frac{P_1 V_1}{N_1 T_1} = \frac{P_2 V_2}{N_2 T_2}$
- The conclusion for the above relation is that it is satisfied by all gases at high temperatures and low pressure.

Ideal Gas

- At all pressure and temperature, a gas exactly fulfils the perfect gas equation.
- The concept of an ideal gas is purely theoretical.
- There is no such thing as an ideal gas. The term "Ideal gas" refers to an ideal gas.
- For low pressure and high temperature, real gases approach ideal gas behaviour.

► Ideal Gas Equation - Equation of State

$$PV = nRT$$

Pressure (atm) $\swarrow$ Volume of gas (l) $\searrow$ moles $\rightarrow$ universal constant. $\rightarrow$ Temp. in K (absolute)

(0.082 atm-l/molK)

$$1 \text{ atm} \approx 1 \text{ bar} \approx 10^5 \text{ Pa} \approx \frac{10^5 \text{ N}}{\text{m}^2} \approx 760 \text{ mm-Hg} \approx 760 \text{ torr.}$$

$$1 \text{ L} = 10^{-3} \text{ m}^3 = 10^3 \text{ cm}^3 = 1 \text{ dm}^3$$

OR

$$PV = k_B N T$$

Boltzmann's constant ($k_B = 1.38 \times 10^{-23} \text{ J/K}$)

$N = \text{No of molecules.}$

→ Density of Gas ($d = \frac{M}{V}$)

Ideal gas eqn, $PV = nRT$

$$PV = \frac{m}{M} RT \quad \left[\begin{array}{l} m = \text{molecular mass} \\ M = \text{Molar mass} \end{array} \right]$$

$$\frac{m}{V} = \frac{PM}{RT}$$

$$d = \frac{PM}{RT}$$

$R \rightarrow \text{SI unit } (8.314 \text{ J/molK})$

$P \rightarrow \text{N/m}^2$

$M \rightarrow \text{in kg.}$

$T \rightarrow \text{in Kelvin}$

$$R = 8.314 \text{ J/molK}$$

$$R = 0.082 \text{ atmL/molK}$$

$$R = 1.98 \text{ cal/mol}^\circ\text{C}$$

$$d = \frac{P \times m \times N_A}{RT}$$

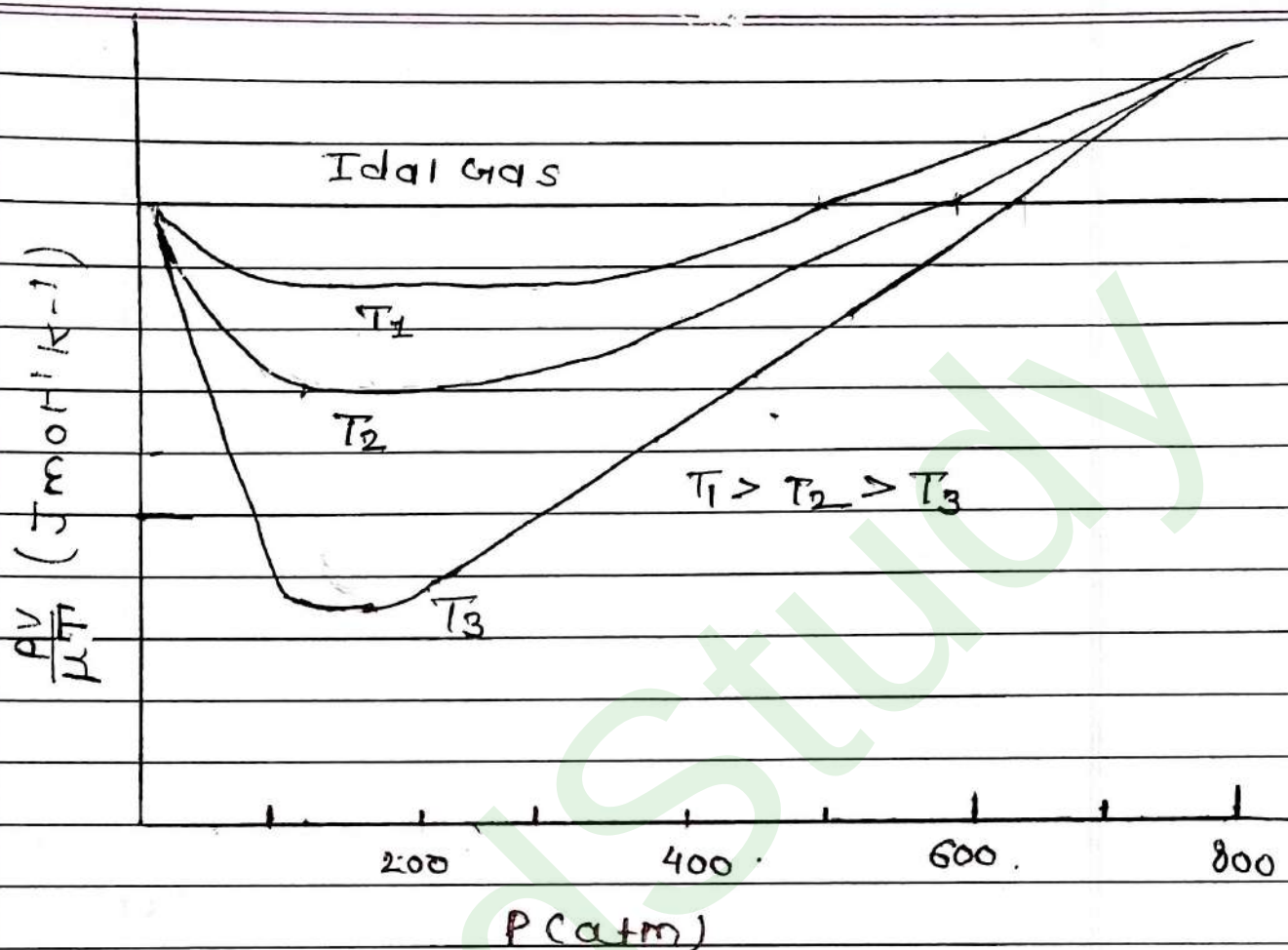
$$= \frac{Pm}{(R)T}$$

$$N_A = k$$

$$\Rightarrow d = \frac{Pm}{kT}$$

Real Gases Deviation from ideal Gas

- For low pressure and high temperatures, real gases approach ideal gas behaviour.
- According to the ideal gas equation:
 $PV = \mu RT$, $\frac{PV}{RT} = \text{constant}$. So for an ideal gas the graph should be a straight line (parallel to the x-axis). This means it has the same value regardless of temperature or pressure.
- However, at high temperature and low pressure, the graph of real gases approaches ideal gas behaviour.
- molecules are far apart at high temperatures and low pressure. when the temperature is raised, the molecules will randomly move away from one another.
- As a result of the reduced molecule interaction, the gas acts like an Ideal gas. when the molecular constituents of the gas do not interact with one another, the optimum behaviour emerges.

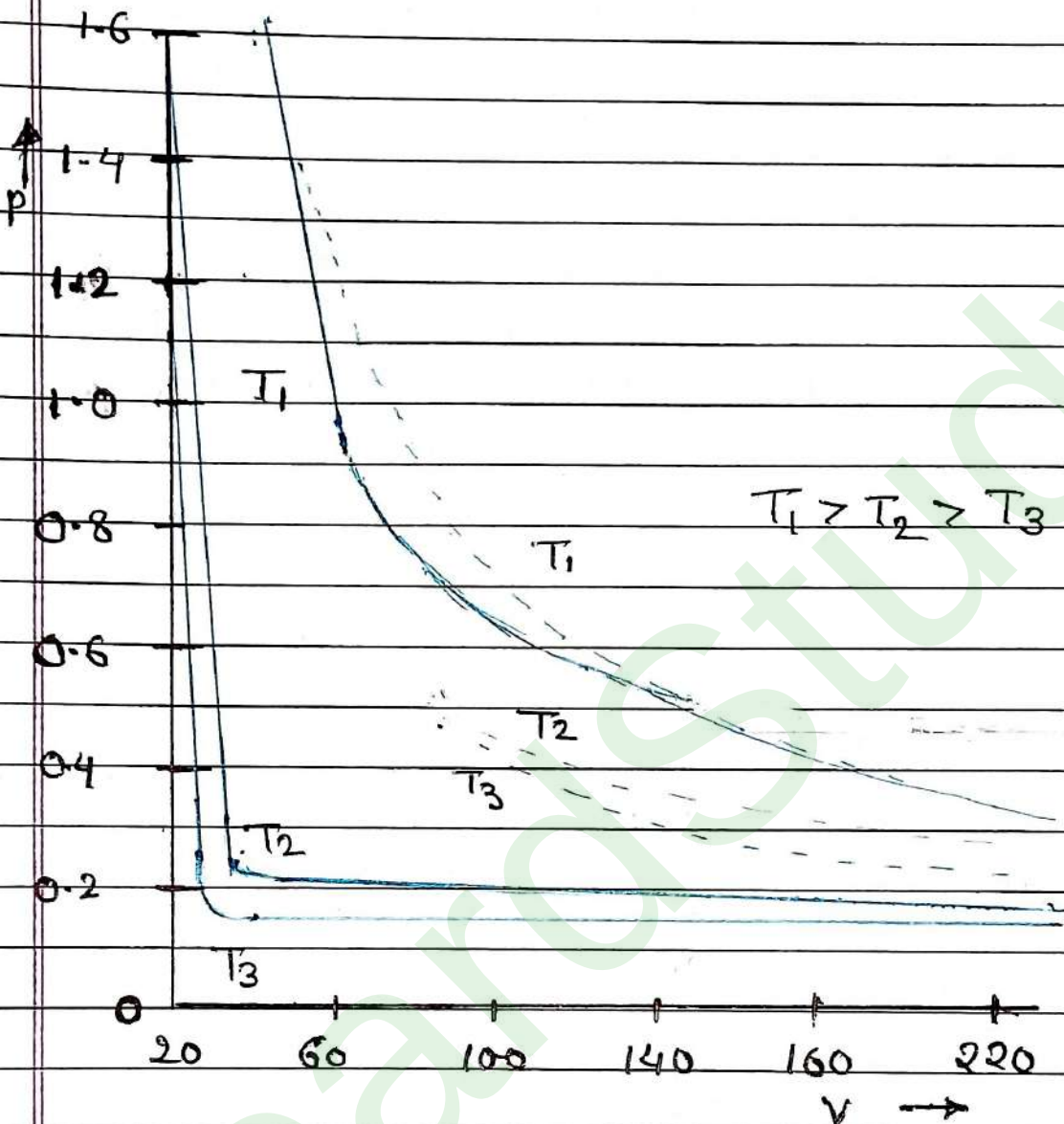


Real gases approach ideal gas behaviour at low pressure and high temperatures.

Deduction of Boyle's law and Charles Law from perfect gas Equation.

- **Boyle's law:** Boyle's law is derived from the perfect gas equation, $PV = \mu RT$. The T is the temperature and μ is the number of moles and both of these value are considered constant.

Therefore, $PV = \text{constant}$. So, at a constant temperature, the pressure of a given mass of gas varies inversely with volume, according to Boyle's law.



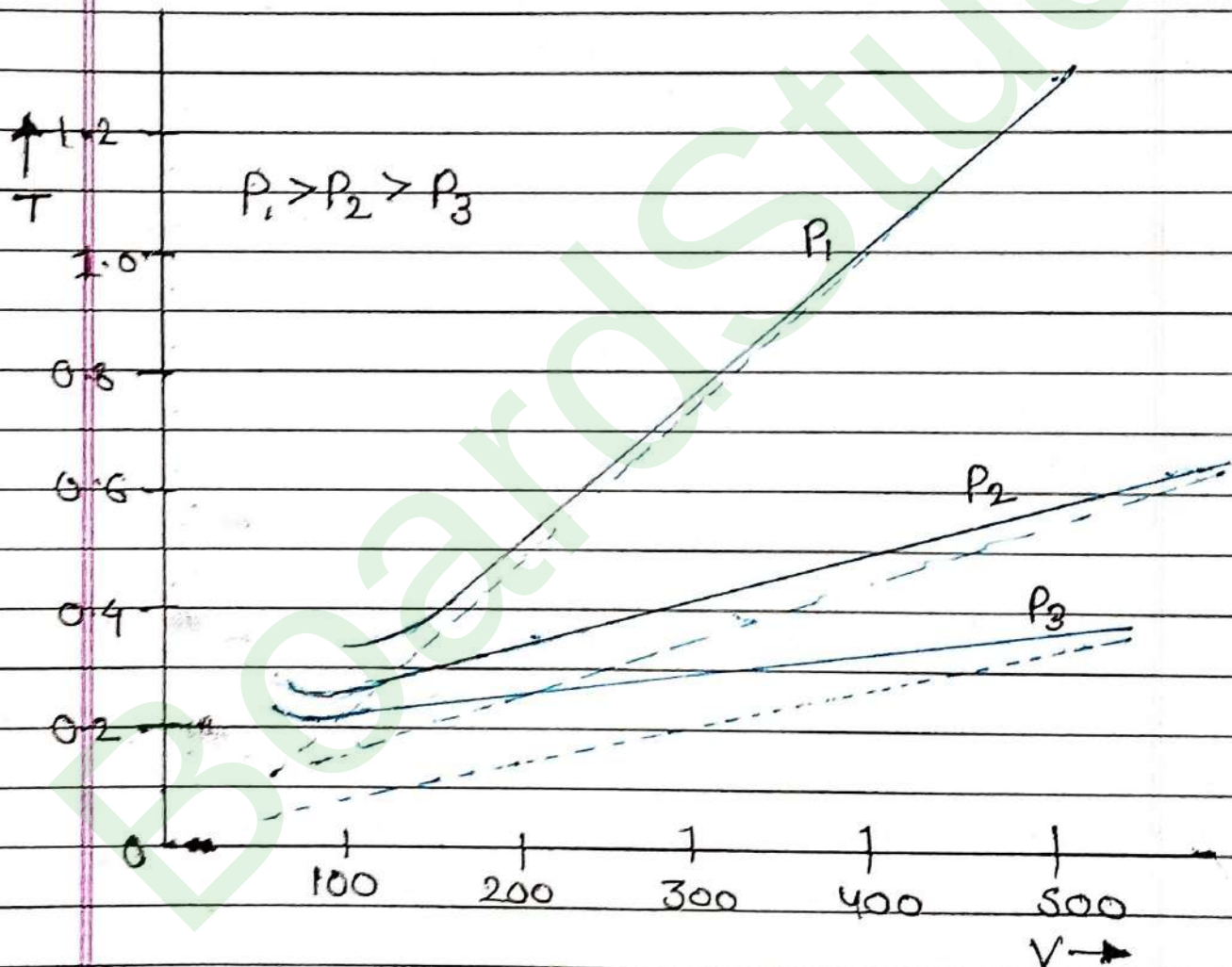
Experimental P-v curves (solid lines) for steam at three temperatures compared with Boyle's law (dotted lines) P in unit of 22 atoms and v in unit of 0.09 litres.

- Charles law:- Here the pressure (P) is considered as constant. The according to the ideal gas equation $PV = \mu RT$.

$$\frac{V}{T} = \frac{\mu R}{p} = \text{constant.}$$

There Fore, $\frac{V}{T} = \text{constant}$

Hence, according to charle's law, at constant pressure, the volume of a gas is directly proportional to its absolute temperature.



Experimental T-V curves (solid lines) for CO_2 at three pressures compared with Charles' law (dotted lines). T is in units of 300K and V in unit of 0.13 litres.

► Kinetic theory of Ideal Gas (Postulates)

- (i) Gas is made up of atoms and molecules.
- (ii) Molecules of same gases are identical in all respects.
- (iii) molecules are constantly in random motion along straight line (only translational motion), (DOF (f) = 3)
- (iv) All the collisions of gas molecules among themselves and wall of container is elastic in nature.
$$\left[\begin{array}{l} \vec{p}_i = -\vec{p}_f \\ KE_i = KE_f \end{array} \right]$$
- (v) The pressure of gas is due to collisions of molecules with wall of container
($P \propto$ no. of collision of molecules per unit Area)
- (vi) The KE of gas depends only and only upon temp. (absolute) and doesn't depend on nature of gas.
($KE \propto T$)
- (vii) The actual volume of gas molecules is negligible when compared to volume occupied by gas.
- (viii) There is no ^{inter} molecular force of attraction among molecules.
- (ix) Gravity is neglected.

(x) molecules of Gas are tiny rigid spheres of negligible volume.

(xi) At Absolute zero, molecular motion stops

→ A gas exerts pressure due to the continuous collisions of its molecules with the walls of the vessel.

Kinetic Gas Equation

$$P = \frac{1}{3} \frac{m N u_{rms}^2}{V}$$

$V \rightarrow$ Volume of gas

$u_{rms} \rightarrow$ Root mean square

$P \rightarrow$ Pressure of gas

Velocity of Gas

$m \rightarrow$ Mass of 1 molecule of gas

molecule.

$N \rightarrow$ no. of molecule of gas.

This can be also written as $\left[P = \frac{1}{3} \rho u_{rms}^2 \right]$

Relation b/w P and KE per unit volume:

$$P = \frac{1}{3} \frac{m N u^2}{V}$$

For 1 mole of gas ($N = N_A$)

$$P = \frac{1}{3} \frac{m N_A u_{rms}^2}{V} = \frac{1}{3} \frac{M u_{rms}^2}{V}$$

$\downarrow$
mass of 1 molecule

$$\Rightarrow PV = \frac{1}{3} M u_{rms}^2 = \frac{1}{3} \times 2 \left[\frac{1}{2} M u_{rms}^2 \right] \text{ (multiply \& divide by 2)}$$

$$\Rightarrow PV = \frac{2}{3} KE$$

3 (1mole)

(kinetic interpretation of T)

$KE = \frac{3}{2} PV$ (1mole)	$PV = nRT$ $PV = RT$	$KE = \frac{3}{2} RT$ (1mole)	$\{ KE \text{ of gas } T \}$
----------------------------------	-------------------------	----------------------------------	------------------------------

For n mole

$$KE = \frac{3}{2} nRT$$

For 1 molecule

$$KE_{1 \text{ molecule}} = \frac{KE (1 \text{ mole})}{N_A}$$

$$KE = \frac{3}{2} kT$$

For ideal gas $f=3$
always.

$\Rightarrow$ Root mean square velocity (U_{rms})

$$U_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + \dots + v_n^2}{n}}$$

RMS velocity for gas molecules

$$U_{rms} \left(\frac{m}{s} \right) = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m}}$$

In terms of P and density -

$$d = \frac{PM}{RT}$$

$$R = 8.314 \text{ J/mol} \cdot K$$

T = in kelvin

M = molar mass in kg

m = mass of 1 molecule
in kg.

$$\frac{RT}{M} = \frac{P}{d}$$

$$U_{rms} = \sqrt{\frac{3RT}{M}}$$

$$U_{rms} = \sqrt{\frac{3P}{d}}$$

► Molecular Velocities

1. Average velocity

$$\left\{ V_{avg} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8KT}{\pi m}} = \sqrt{\frac{8P}{\pi d}} = \frac{v_1 + v_2 + \dots + v_n}{n} \right\}$$

2. Root mean square velocity

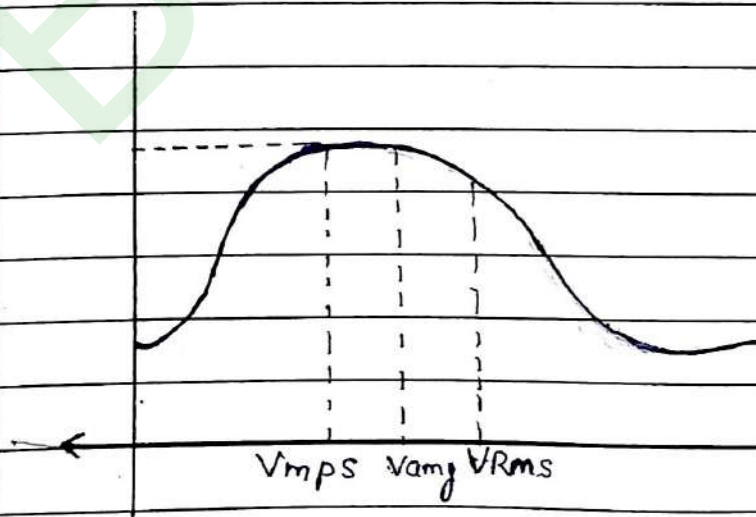
$$\left\{ V_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3KT}{m}} = \sqrt{\frac{3P}{d}} \right\}$$

3. Most Probable velocity

maximum fraction of molecules passes this velocity

$$\left\{ V_{mps} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2KT}{m}} = \sqrt{\frac{2P}{d}} \right\}$$

► Maxwell's Speed Distribution Law



$$V_{RMS} > V_{avg} > V_{mps}$$

$$R > A > M$$

$$\sqrt{\frac{3RT}{M}} > \sqrt{\frac{8RT}{\pi M}} > \sqrt{\frac{2RT}{M}}$$

$$\sqrt{3} > \sqrt{\frac{8}{\pi}} > \sqrt{2}$$

$$\sqrt{3} > \sqrt{2.5} > \sqrt{2}$$

Mean Free Path

- The average distance travelled by molecule between two successive collisions is known as mean free path.

$$\lambda_m = \frac{\lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_n}{n}$$

$$\lambda_m = \frac{1}{\sqrt{2} n \pi d^2}$$

λ_m = mean free path.

n = no. of molecules per unit volume
(number density)

d = diameter of molecule

Law of Equipartition of Energy:

Degree of freedom

- Degree of Freedom are separate displacements or rotations that specify a body's or system's orientation.
- Three coordinates are required to define the location of a molecule that is free to move in space.
- It must move in a plane if it is limited.
- It just needs one coordinate to locate itself if it is confined to moving down a line.
- Consider a room in which a strong rope is tied from one wall to the next.
- Take a straight ball and place it on the rope.
- There is only one degree of freedom for the ball. It can only move in one dimension at a time.
- Consider the situation where the ball is on a two-dimensional floor and can go in two directions.

- There are two degrees of freedom for the ball.
- Consider throwing the ball in three-dimensional space. The ball can then move in three dimensions.
- As a result, the degree of freedom indicates how many different ways a body can move, rotate or vibrate.

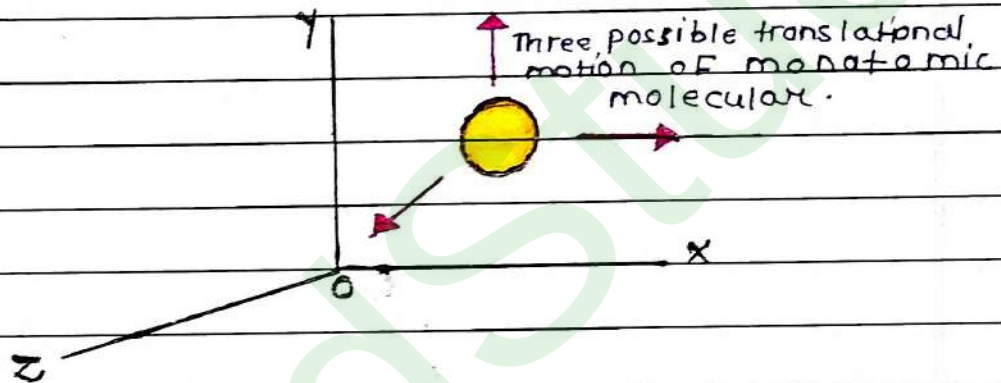
Categories of Degree of freedom

1. Translational Degree of Freedom
2. Rotational Degree of Freedom
3. Vibrational Degree of Freedom.

Translational Degree of freedom

- The term 'translation' refers to the movement of the entire body from one position to another.
- Consider the oxygen molecule, which is made up of two linked oxygen atoms.
- The two oxygen atoms, as well as the link, are referred to as the complete body.

- Translational movement occurs when the entire body moves from one position to another.
- Consider a molecule that is free to move in space and hence requires three coordinates to define its location (x, y and z)



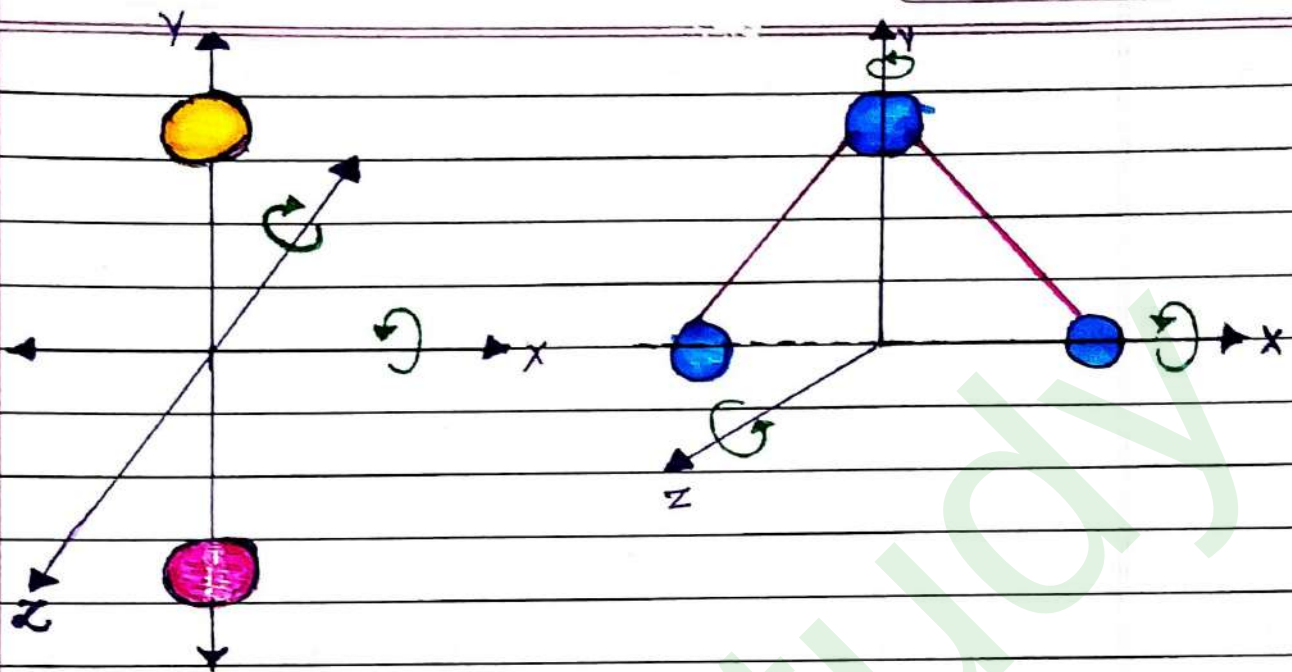
- As a result, it has three degrees of freedom.
- Similarly, a molecule that is free to move on a two dimensional plane requires two coordinates to pinpoint its location.
- As a result, it has two degrees of freedom.
- Only translational degrees of freedom exist in mono-atomic gas molecules. This refers to gases with only one atom.

- For example, a single He atom makes up a Helium atom. It will have translation degrees of Freedom.
- Each degree of Freedom in translation adds a word containing the square of motion Variable.
- The velocity is the motion variable (v_x, v_y, v_z).
- Energy will be contributed by the term $(\frac{1}{2})mv_x^2$
- Kinetic energy is the energy involved in the movement of a molecule from one location to another.
- The average of each of these terms in thermal equilibrium is $(\frac{1}{2})k_B T$.

• Rotational Degree of freedom

- Independent rotations that specify a body's or system's orientation.
- A part of the body rotates about another part of the body.
- Only diatomic gases have rotational degrees of freedom.
- In addition to translational degrees of freedom, diatomic molecules degree of freedom.
- It's Feasible in diatomic compounds because two atoms are linked by a bond. As a result, the rotation of one atom in relation to another atom.
- There is a translational degree of freedom in diatomics, as well as rotational degree of freedom.
- For instance, two oxygen atoms are connected by a bond. Two axes are perpendicular to each other.

- Along the two axes, two rotations are conceivable.
- They have three degrees of freedom in translation and two degrees of freedom in rotation.
- As a result, a component in the energy that contains the square of a rotating variable of motion is contributed by the rotational degree of freedom.
- Angular momentum is the source of the rotational variable of motion.
- V_x, V_y, V_z is the linear velocity. Angular velocity is written as $\omega_x, \omega_y, \omega_z$.
- The three rotational degree of freedom along the two perpendicular axis are:
$$E_R = \left(\frac{1}{2}\right) (I_1 \omega_1^2) + \left(\frac{1}{2}\right) (I_2 \omega_2^2)$$
- The three translational degree of freedom will be $\frac{1}{2} m_x v_x^2$, $\frac{1}{2} m_y v_y^2$ and $\frac{1}{2} m_z v_z^2$.
- The two rotational degrees of freedom are: $\left(\frac{1}{2}\right) (I_1^2 \omega_1^2)$ and $\left(\frac{1}{2}\right) (I_2^2 \omega_2^2)$

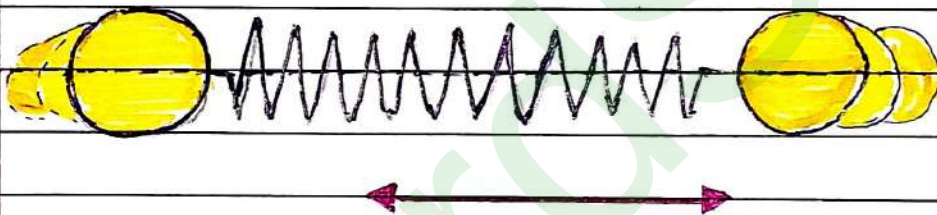


Vibrational Degree of Freedom

- Some molecules exhibit a mode of vibration, in which the atoms oscillate along the inter-atomic axis similar to a one-dimensional oscillator.
- In some molecules, this vibration can be seen.
- Consider the following scenario:
 Co atoms oscillates like a one-dimensional oscillator along the inter-atomic axis.
- Consider two atoms that vibrate in the same direction along the inter-atomic axis.

- The vibrational Energy terms include the Square of vibrational motion Variables.
- The total vibrational energy. $E_v = \frac{1}{2} m \left(\frac{dy}{dt} \right)^2 + \frac{1}{2} k y^2$, where, $\frac{1}{2} m \left(\frac{dy}{dt} \right)^2$ is the kinetic energy, $\frac{1}{2} k y^2$ is the potential energy.
- Two terms are contributed by vibrational degree of freedom.

vibrational motion along the molecular axis



Molecular axis

Comparison Between three Energy Modes

Translational	Rotational	Vibrational
$E_t = \frac{1}{2} m v_x^2$	$E_r = \frac{1}{2} I \omega^2$	$E_v = \frac{1}{2} m \left(\frac{dy}{dt} \right)^2 + \frac{1}{2} k y^2$
one squared term is depicted in the above expression	one squared term is depicted in the above expression	Two squared terms are being depicted in the above expression

Law of Equipartition of Energy

The total energy is equally distributed across all conceivable energy modes in equilibrium, according to this law, with each mode having an average energy equal to $\frac{1}{2} k_B T$.

- $\frac{1}{2} k_B$ is contributed by each translational degree of Freedom.
- Each degree of rotational Freedom provides $\frac{1}{2} k_B$.
- Each degree of vibrational Freedom provides two times of $\frac{1}{2} k_B$.

→ Specific Heat Capacity for monoatomic Gases :-

- mono-atomic gases will only have one degree of freedom which is the translation.
- They can only have three translational degree of Freedom at most.
- Each degree of freedom will add $\frac{1}{2} k$ to the total.

- As a result, three degree of freedom will each contribute $\frac{3}{2} k$.
- The total internal energy of 1 mole of Gas U is calculated using the law of energy equipartition.

$$U = \frac{3}{2} k_B T N_A = \frac{3}{2} RT$$

- At constant volume, the specific heat capacity will be: $C_v = \frac{dU}{dT} = \frac{3}{2} R$

- For an Ideal gas $C_p - C_v = R$.
Now $C_p = \frac{5}{2} R$

- The ratio of specific heat will now be: $\gamma = \frac{C_p}{C_v} = \frac{5}{3}$

Specific Heat of Diatomic Gases (Rigid)

- A stiff diatomic gas has translational and rotational degrees of Freedom but no vibrational degree of Freedom.
- They are oscillators that are stiff.
- There are three translational degrees of Freedom and two rotational degrees of Freedom in a stiff diatomic molecule. There a total of 5 degrees of Freedom.
- Each degree of Freedom will contribute $\frac{1}{2} k_B$ according to the law of energy equipartition.
- As a result, the Fifth degree of Freedom will add $\frac{5}{2} k_B$ to the total.
- The total internal energy for one mole of gas will be $U = \frac{5}{2} k_B T N_A = \frac{5}{2} RT$.
- At constant volume the specific heat capacity will be $C_v = \frac{dU}{dT} = \frac{5}{2} R$.

- For a rigid diatomic, the specific heat capacity at constant pressure will be given as:

$$C_p = \frac{7}{2} R$$

- The ratios of specific heat will be

$$\gamma = \frac{C_p}{C_v} = \frac{7}{5}$$

* Specific Heat of diatomic Gases (non-rigid)

- Translational, rotational, and vibrational degree of freedom exist in a non-rigid diatomic gas.
- There will be three degree of freedom in translation, two degrees of freedom in rotation, and one degree of freedom in vibration.
- The total contribution by translational is $\frac{1}{2} k_B T$, by rotational is two times of translational and vibrational is k_B .
- Total internal energy for one mole is equal to $\frac{5}{2} k_B T + k_B T =$

$$\frac{7}{2} k_B T = \frac{7}{2} RT$$

★ Specific Heat for Polyatomic molecules

- Polyatomic gases will have three degrees of freedom in translation, three of freedom in rotation, and a 'f' number of vibrational modes.

- For one mole of the gas the total internal energy will be:

$$\left(\frac{3}{2} k_B T + \frac{3}{2} k_B T + F k_B T\right) \times N_A = (3+F) RT$$

- $C_v = \frac{dU}{dT} = (3+F) R$

- $C_p = C_v + R = (4+F) R$

- $\gamma = \frac{C_p}{C_v} = \frac{(4+F)}{(3+F)}$

Specific Heat capacity for solids

- Consider the number of atoms in a solid, which is N . Each atom can fluctuate around its mean position.
- The degree of freedom for vibrational motion will $k_B T$.
- The average energy for one-dimensional is $k_B T$ and for three-dimensional will be $3k_B T$.
- The total internal energy for one mole of solid will be $3k_B T N_A = 3RT$.
- The volume change is very less in solids at constant pressure.
 $\Delta Q = \Delta U + P \Delta V$. So, $\Delta V = 0$
therefore $\Delta Q = \Delta U$.
- $C_v = \left(\frac{dU}{dT} \right)_v$
- $C_p = \left(\frac{dQ}{dT} \right)_p$ as $\Delta Q = \Delta U$, therefore $C_v = \frac{dU}{dT} = 3R$.
- Therefore, $C_p = C_v = 3R$.

Specific Heat Capacity of water

- If water is considered solid, then there will be an 'N' number of atoms.
- So, the Average energy of atom is $3k_B T$
- Therefore there are three molecules in a water molecule (H_2O)
- So, the total internal energy will be $U = 3k_B T \times 3 \times N_A = 9RT$.
- $C_p = C_v = 9R$.